SESSION 10

Root Cause Analysis (RCA)

- Event Correlation
- Causal Analysis
- Topology-Aware RCA

Agenda

Session 10 — Root Cause Analysis

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02 RCA in Network Management Context

03 Event Correlation: Concepts & Types

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What is Root Cause Analysis (RCA)?

Root Cause Analysis (RCA) is a systematic process for identifying the fundamental source of faults or failures in complex systems — going beyond symptoms to uncover WHY a problem occurred so it can be permanently resolved.

DETECT

Identify abnormal events and alarms in the network

DIAGNOSE

Determine the root cause from correlated evidence

RESOLVE

Apply targeted fixes to prevent recurrence

KEY BENEFITS

- ↓ 60–80% MTTR Reduction
- ↓ 90% Alarm Noise
- ↑ 85–95% Accuracy
- ↓ 55% Repeat Incidents

RCA in Network Management

Why traditional approaches fail at scale

Scale

Modern networks have 10K–1M+ devices generating millions of events per hour

Complexity

Multi-vendor, multi-layer topologies with intricate interdependencies

Speed

MTTR requirements demand root cause identification in seconds, not hours

Noise

Up to 99% of alarms may be symptomatic — masking the true root cause

Industry Fact: A large telco generates ~2.5 million alarms/day — human analysts can review fewer than 0.1%

Event Correlation: Overview & Types

Transforming raw alarms into actionable insights

Temporal

Events correlated by time windows — alarms within T seconds are related

Spatial

Events correlated by network proximity and topology adjacency

Rule-Based

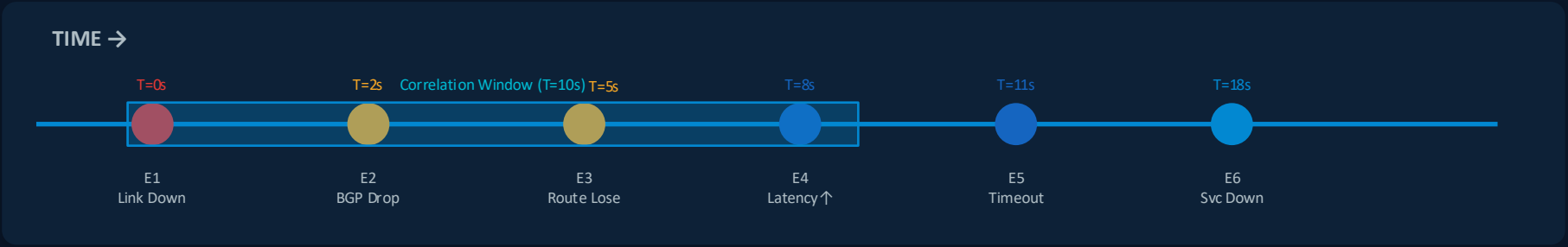
Expert knowledge encoded as IF-THEN correlation rules

Statistical/ML

Pattern learning and probabilistic models from historical data

Temporal Correlation

Time-window based event grouping and analysis



Sliding Window

Fixed-size time window moves along event stream; events within window are candidates for correlation. Window size W is a critical parameter.

Time-Decay Model

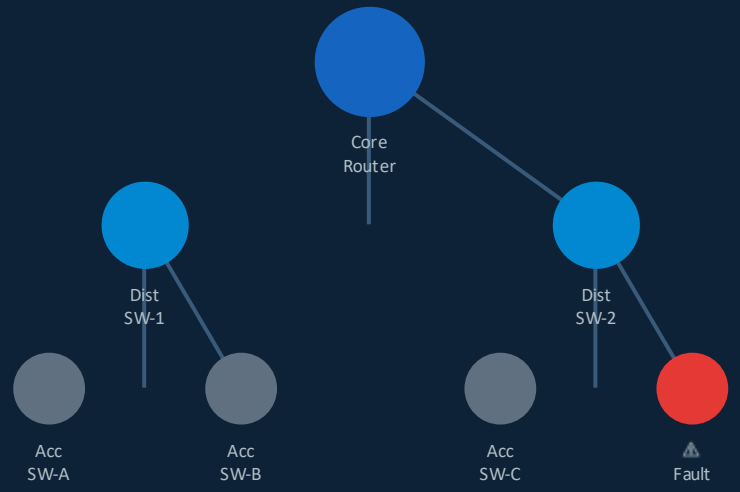
Event relevance decreases exponentially over time. Recent events have higher correlation weight than older ones — mimics causal proximity.

Sequence Mining

Discovers frequent temporal patterns (A→B→C within T seconds) using algorithms like GSP, PrefixSpan on historical alarm logs.

Spatial & Rule-Based Correlation

SPATIAL CORRELATION



Spatial proximity propagates alarms up/down topology

RULE-BASED CORRELATION

Rule 1

```
IF  "Link Down" (Device X)
AND "BGP Session Drop" (Peer Y)
WITHIN  30s
THEN  Root Cause = Physical Link Failure
```

Rule 2

```
IF  "High CPU" (Router A)
AND "Packet Loss > 5%"
AND "Memory > 90%"
THEN  Root Cause = Resource Exhaustion
```

Rule 3

```
IF  "Power Alarm" (Site Z)
AND count(Down events) > 5
WITHIN  60s
THEN  Root Cause = Power Outage
```

Rule sources: Expert knowledge, SNMP MIBs, vendor advisories

Statistical & ML-Based Correlation

Learning patterns from historical alarm data

Probabilistic Models

P(B|A) — compute conditional alarm probabilities from historical co-occurrence matrices. High P(B|A) indicates causal link A→B

Clustering Algorithms

K-means, DBSCAN, hierarchical clustering group similar alarms. Alarms in same cluster likely share root cause or propagation path

Anomaly Detection

Isolation Forest, Autoencoders flag deviations from baseline. Anomalous alarm patterns trigger deeper RCA investigation

Association Rule Mining

Apriori, FP-Growth mine frequent alarm itemsets. High-confidence rules (e.g., {A,B}→C at 94%) encode latent correlations

Causal Analysis Methodologies

Systematic techniques to trace fault causality

Physical Fault → Link Failure → Protocol Event → Service Degradation → User Impact

5-Why Analysis

Iteratively asks 'Why?' 5 times. Each answer reveals the next layer of causality. Simple, non-technical, widely applicable.

Fault Tree Analysis

Top-down deductive logic tree. AND/OR gates model how component failures combine to produce system-level faults (FTA).

Ishikawa / Fishbone

Visualizes cause-effect relationships. 6M categories: Machine, Method, Man, Material, Measurement, Mother Nature (Environment).

Bayesian Causal Networks

Probabilistic directed acyclic graphs. P(Fault|Evidence) computed via Bayes theorem — handles uncertain, incomplete observations.

Causal Inference (DoWhy)

Statistical causal reasoning using do-calculus. Separates correlation from causation. Supports counterfactual 'what-if' analysis.

RCFA (Root Cause Failure)

Engineering-oriented analysis combining physical evidence, operating data, and failure mode analysis for hardware faults.

5-Why Analysis

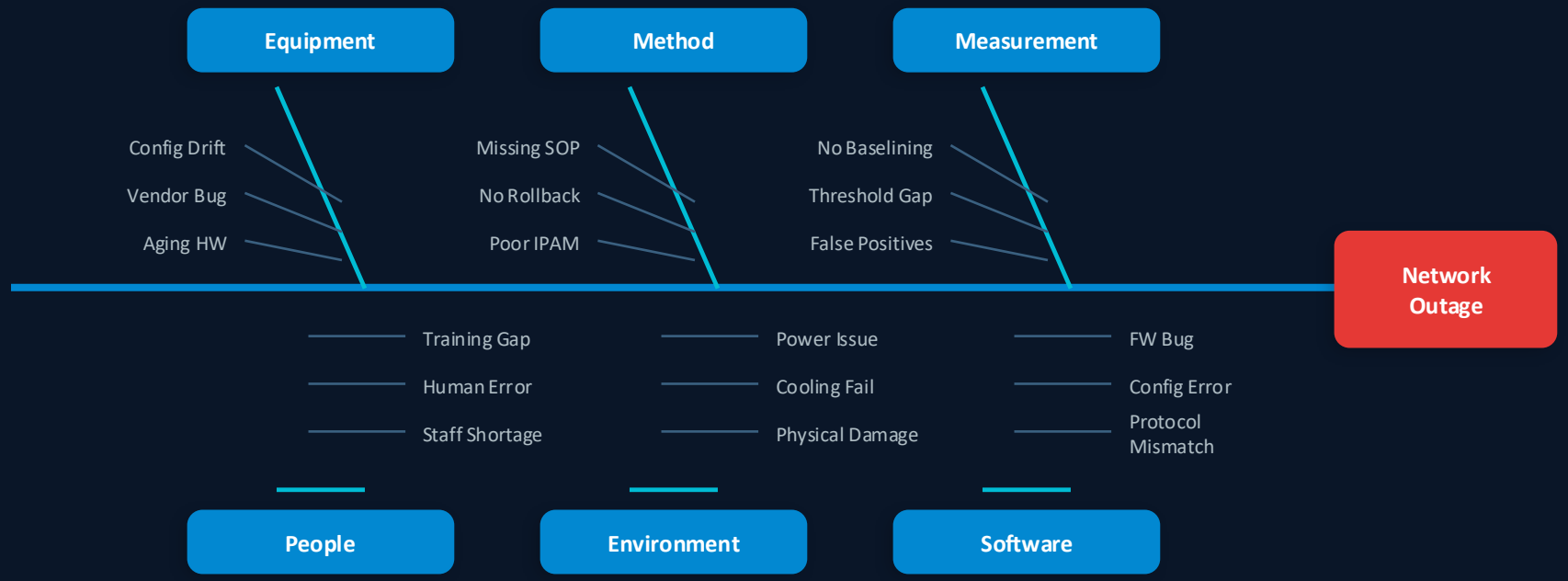
Case: Network service degradation at 14:32 UTC

Problem	Users cannot access the application → **Web server returning 504 Gateway Timeout**
Why 1	Why is the web server returning 504? → **Load balancer cannot reach backend servers**
Why 2	Why can't the load balancer reach backends? → **Core switch port GE0/1 is down**
Why 3	Why is switch port GE0/1 down? → **LACP negotiation failed on the port channel**
Why 4	Why did LACP negotiation fail? → **Firmware update changed LACP timer defaults**
Why 5	Why were LACP timers changed by firmware update? → **Change management did not include LACP validation**

✅ **ROOT CAUSE: Incomplete change management process for firmware updates — LACP validation step missing from runbook**

Ishikawa / Fishbone Diagram

Root cause visualization across six cause categories



Fault Propagation Algorithm

Computing root cause and impact using graph traversal

```
ALGORITHM: TopologyRCA(G, A)

INPUT:        Graph G=(V,E), Active Alarm Set A
INIT:         candidates ← {v ∈ V : alarm(v) = true}
FOR           each candidate v in A:
  BFS/DFS     from v on G (neighbors)
  COMPUTE     propagation score P(v)
  IF          score(v) > threshold θ:
  →           add v to rootCauses
SUPPRESS      downstream symptomatic alarms
RANK          rootCauses by P(v) descending
OUTPUT:       ranked root cause list
```

Time Complexity: O(|V| + |E|) — linear in graph size

O(V+E) Time Complexity

94% Localization Accuracy

<2s Detection Latency

90%↓ Alarm Noise Reduction

Probabilistic & Bayesian RCA

Reasoning under uncertainty with conditional probabilities

Bayes Theorem for RCA:

$$P(Cause \mid Evidence) = [\, P(Evidence \mid Cause) \times P(Cause) \,] / P(Evidence)$$

1

Build BN

Define nodes (faults, symptoms) and directed edges (causal relationships) from expert knowledge or learned from data

2

Set Priors

Assign prior probabilities P(Cause) based on historical failure rates and MTBF data for each component

3

Observe

Collect active alarms as observed evidence. Instantiate evidence nodes in the network (E=true)

4

Infer

Run belief propagation / variable elimination to compute P(RootCause | E). Highest posterior = most likely root cause

Machine Learning Approaches for RCA

State-of-the-art ML models for automated fault diagnosis

Random Forest

Ensemble — Supervised

Accurac
y: 92%

Trains on labeled alarm-root cause pairs. Feature importance identifies key diagnostic signals. Fast inference: ~10ms per classification. Handles class imbalance well.

LSTM Networks

Deep Learning — Temporal

Accurac
y: 88%

Processes alarm sequences as time series. Captures long-range temporal dependencies in alarm streams. BiLSTM variant improves correlation accuracy by ~8%.

Graph Neural Networks

GNN — Topology-Aware

Accurac
y: 94%

Embeds network topology directly into model. Message passing propagates fault signals along graph edges. State-of-the-art for topology-aware RCA (Deng et al., 2021).

Autoencoders

Unsupervised — Anomaly

Accurac
y: 85%

Learns normal alarm pattern encoding. Reconstruction error detects novel fault signatures. Requires no labeled data — suitable for zero-day fault patterns.

End-to-End RCA Process Workflow

Integrated pipeline from detection to prevention

1 DETECT

Real-time alarm ingestion from SNMP, Syslog, NETCONF, streaming telemetry

2 CORRELATE

Apply temporal, spatial, rule-based and ML correlation engines

3 ISOLATE

Suppress symptom alarms, identify candidate root cause devices

4 DIAGNOSE

Probabilistic RCA scoring, topology traversal, 5-Why analysis

5 REMEDIATE

Automated or guided remediation: rollback, reroute, restart, escalate

6 PREVENT

Update knowledge base, retrain models, revise runbooks and change processes

↺ Continuous Feedback Loop: Resolved incidents enrich training data → improved correlation accuracy over time

RCA Tools & AIOps Platforms

Commercial and open-source solutions for automated RCA

IBM Netcool / OMNIbus

Enterprise NMS

★★★★☆

Industry leader for telco/enterprise RCA. Event correlation, topology mapping, CEMM. Integrates with IBM Watson AIOps for ML-enhanced RCA.

Moogsoft AIOps

AI-Driven Correlation

★★★★★

Purpose-built AIOps platform. Situation clustering, noise reduction (90%+), causation inference. REST API for ITSM integration.

Splunk ITSI

IT Analytics

★★★★☆

Service-centric monitoring with Episode Review and Adaptive Thresholding. Glass tables for real-time topology-aware dashboards.

PagerDuty AIOps

Incident Management

★★★★☆

Alert grouping, intelligent triage, past incidents ML matching. Event Intelligence module for automated RCA suggestions.

Dynatrace Davis AI

Full-Stack Observability

★★★★★

Automatic full-stack dependency mapping. Davis AI causation engine correlates across infrastructure, APM, and user sessions.

ServiceNow ITOM

Service Management

★★★☆☆

Discovery, Service Mapping and Event Management integrated. AIOps connector for third-party ML alarm correlation.

Ref: Gartner Magic Quadrant for AIOps (2023); Forrester Wave: AIOps Q1 2023; Moogsoft.com; dynatrace.com/aiops

RCA Metrics & KPIs

Measuring the effectiveness of automated root cause analysis

60-80%

MTTR Reduction

Mean Time To Repair

~90%

Alarm Noise Reduction

Symptomatic alarms suppressed

55%↓

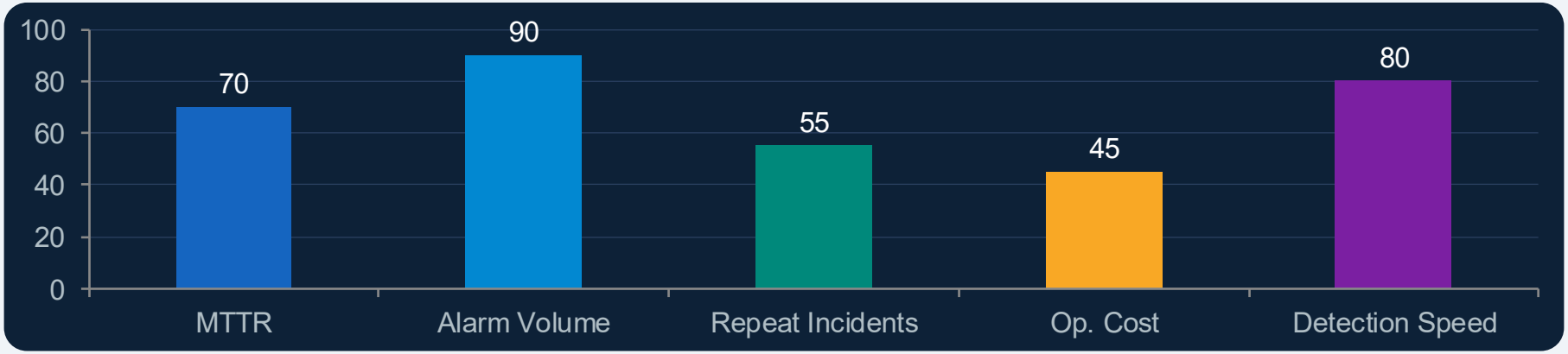
Repeat Incidents

Permanent resolution rate

85-95%

RCA Accuracy

Correct root cause identified



Challenges & Limitations

Known obstacles in deploying production RCA systems

Data Quality

Missing, delayed or duplicate alarms degrade correlation accuracy. Requires robust data normalization pipeline.

FIX *Deduplication, normalization layers*

Dynamic Topologies

SDN/NFV environments change topology in real-time, invalidating static graph models within minutes.

FIX *Streaming topology sync via NETCONF*

Labeled Data Scarcity

ML models require labeled root cause datasets, which are expensive and time-consuming to create from incidents.

FIX *Transfer learning, semi-supervised*

Multi-Domain Faults

Faults spanning IP, optical, and cloud layers require cross-domain topology fusion — technically challenging.

FIX *Multi-layer graph abstraction*

Alarm Storms

High-volume alarm bursts can overwhelm correlation engines, causing false positives and missed root causes.

FIX *Rate limiting, priority queuing*

Explainability

Black-box ML models generate root cause predictions without human-interpretable reasoning paths.

FIX *SHAP values, attention visualization*

Best Practices for Effective RCA

Operational guidelines for production deployments

Maintain a CMDB

Keep Configuration Management DB synchronized with real-time topology changes. Stale CMDB is the #1 cause of topology-RCA failures.

Build Historical Baseline

Collect 3–6 months of alarm data before training ML models. Include seasonal patterns and maintenance event labels.

Layer Multiple Techniques

No single method works universally. Combine rule-based + ML + topology for 20–30% accuracy improvement over single approaches.

Close the Learning Loop

Every resolved incident should update the knowledge base. Continuous retraining improves model accuracy 5–10% per quarter.

Test with Chaos Engineering

Inject controlled faults (link failures, CPU spikes) to validate RCA accuracy before production deployment.

Integrate with ITSM Workflows

RCA findings should automatically create/update JIRA, ServiceNow, or PagerDuty tickets to close the loop between detection and resolution.

SESSION 10 — KEY TAKEAWAYS

Event Correlation is the foundation of RCA — temporal, spatial, rule-based and ML approaches work best in combination, reducing alarm noise by up to 90%.

Causal Analysis (5-Why, FTA, Bayesian Networks) provides the analytical framework to move from symptoms to root cause systematically.

Topology-Aware RCA leverages network graph structure (G=V,E) to propagate fault signals and suppress downstream symptomatic alarms.

ML approaches (GNN, LSTM, Random Forest) achieve 85-94% RCA accuracy but require quality labeled data and continuous retraining.

Effective RCA systems combine automated detection, correlation, topology analysis and ITSM integration — reducing MTTR by 60-80%.